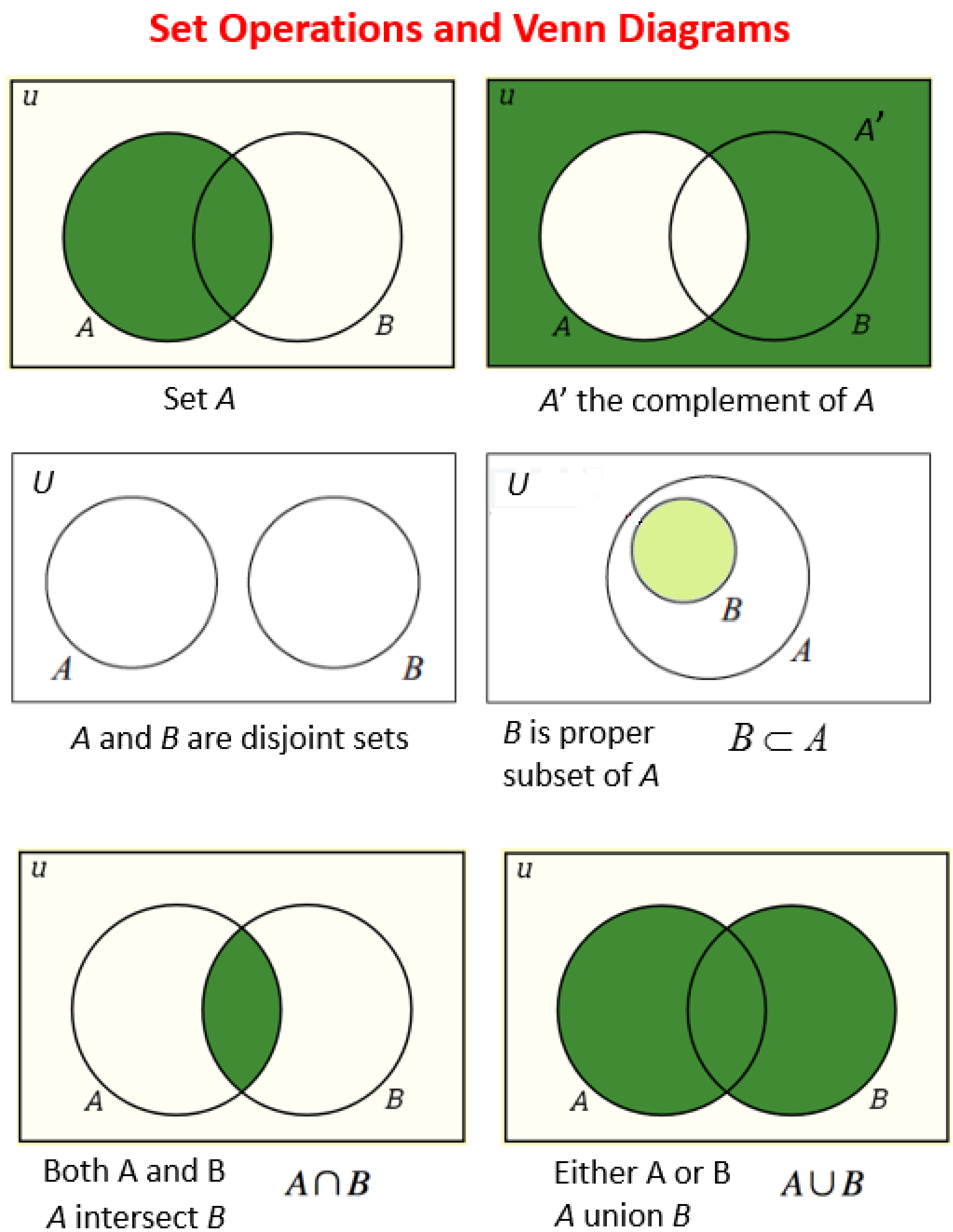
**What is a Venn Diagram?**
A Venn Diagram is a pictorial representation of the relationships between sets.

We can represent sets using **Venn diagrams.** In a Venn diagram, the sets are represented by shapes; usually circles or ovals. The elements of a set are labeled within the circle.

The following diagrams show the set operations and Venn Diagrams for Complement of a Set, Disjoint Sets, Subsets, Intersection and Union of Sets. Scroll down the page for more examples and solutions.



The set of all elements being considered is called the **Universal Set (U)** and is represented by a rectangle.

The **complement of a set A is denoted by A'**, is the set of elements in U but not in A.
$A' = \{x | x \in U \text{ and } x \notin A\}$

Set A and B are **disjoint sets** if they do not share any common elements.

B is a **proper subset** of A. This means B is a subset of A, but $B \neq A$.

The **intersection of A and B** is the set of elements in both set A and set B. $A \cap B = \{x | x \in A \text{ and } x \in B\}$
The **union of A and B** is the set of elements in set A or set B. $A \cap B = \{x | x \in A \text{ or } x \in B\}$

$A \cap \emptyset = \emptyset$

$A \cup \emptyset = A$

**Set Operations and Venn Diagrams**

**Example:**

1. Create a Venn Diagram to show the relationship among the sets.

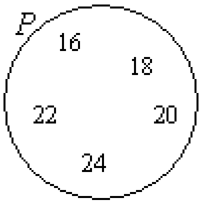
   U is the set of whole numbers from 1 to 15.
   A is the set of multiples of 3.
   B is the set of primes.
   C is the set of odd numbers.

2. Given the following Venn Diagram determine each of the following set.
   *a*) $A \cap B$
   *b*) $A \cup B$
   *c*) $(A \cup B)'$
   *d*) $A' \cap B$
   *e*) $A \cup B'$

**Example:**

Given the set *P* is the set of even numbers between 15 and 25. Draw and label a Venn diagram to represent the set *P* and indicate all the elements of set *P* in the Venn diagram.

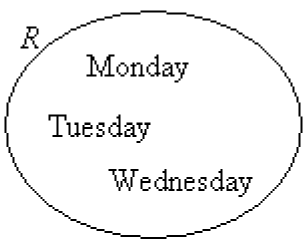
**Solution:**
List out the elements of *P*.
$P = \{16, 18, 20, 22, 24\}$ ← ‘between’ does not include 15 and 25
Draw a circle or oval. Label it *P* . Put the elements in *P*.




**Example:**

Draw and label a Venn diagram to represent the set
$R$ = {Monday, Tuesday, Wednesday}.

**Solution:**
Draw a circle or oval. Label it $R$. Put the elements in $R$.

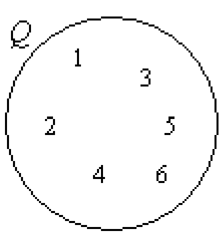



**Example:**

Given the set $Q = \{x: 2x - 3 < 11, x \text{ is a positive integer}\}$. Draw and label a Venn diagram to represent the set $Q$.

**Solution:**

Since an equation is given, we need to first solve for $x$.
$2x - 3 < 11 \Rightarrow 2x < 14 \Rightarrow x < 7$




So, $Q = \{1, 2, 3, 4, 5, 6\}$
Draw a circle or oval. Label it $Q$.
Put the elements in $Q$.